

Electricity Demand Forecasting for Germany: A Time Series Case Study

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Github: <https://github.com/indhu0204/electricity-demand-forecasting>

1. Introduction

Electricity demand forecasting is critical for power system planning, market operations and reserve scheduling because forecast errors directly affect system reliability and operating costs. Accurate load forecasts allow operators to balance supply and demand, schedule generation efficiently and prepare for periods of unusually high demand.

This study evaluates several forecasting approaches for weekly German electricity demand, starting from simple statistical benchmarks and progressing to more advanced time-series and machine learning models. The aim is not only to identify the most accurate model, but to explain why some methods perform better or worse on this series, recognising that complex models do not automatically outperform simpler ones when the data have strong and stable seasonality. Advanced models are therefore compared against credible benchmarks, and the results are interpreted in relation to both the characteristics of the data and the modelling choices made in code.

The dataset consists of weekly German demand from 4 January 2015 to 4 October 2020, drawn from the Open Power System Data platform. A training period from 2015-01-04 to 2018-10-07 and a test period from 2018-10-14 to 2020-10-04 are used so that all models are evaluated on unseen data in a realistic forecasting setting.

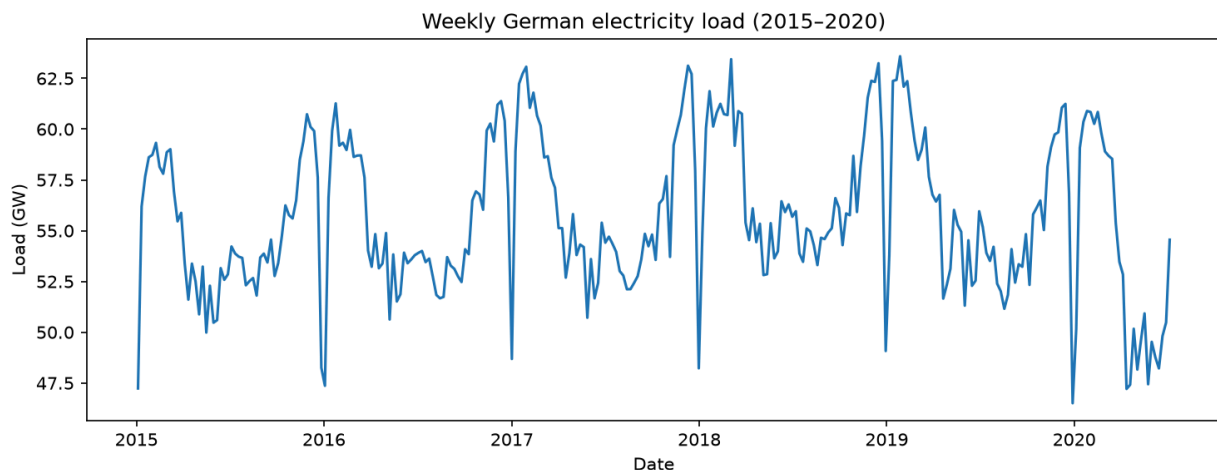


Figure 1. Full weekly electricity demand series with the train/test split marked.

2. Data and forecasting problem

The data form a weekly univariate time series of German electricity demand measured in gigawatts, indexed by date, with 301 observations from 2015–2020, of which 197 weeks are used for training and 104 for testing. Visual inspection shows pronounced annual seasonality with recurring winter peaks and lower summer demand, implying that seasonal structure is a major source of predictability in this series.

The forecasting task is defined as one-week-ahead prediction over the entire 104-week test period, using only information available up to the forecast origin and, where relevant, lagged or engineered features. This design avoids information leakage and ensures that reported metrics reflect genuine out-of-sample performance rather than in-sample fit. Accuracy is assessed using MAE, RMSE, MASE and bias: MAE measures typical error size, RMSE penalises large misses more strongly, MASE scales performance relative to a naive benchmark, and bias indicates systematic over- or under-prediction. When exogenous variables such as temperature or calendar indicators are included, they are motivated by the fact that electricity demand is strongly shaped by weather and recurring calendar effects, which can explain departures from the basic seasonal pattern that purely univariate models can miss.

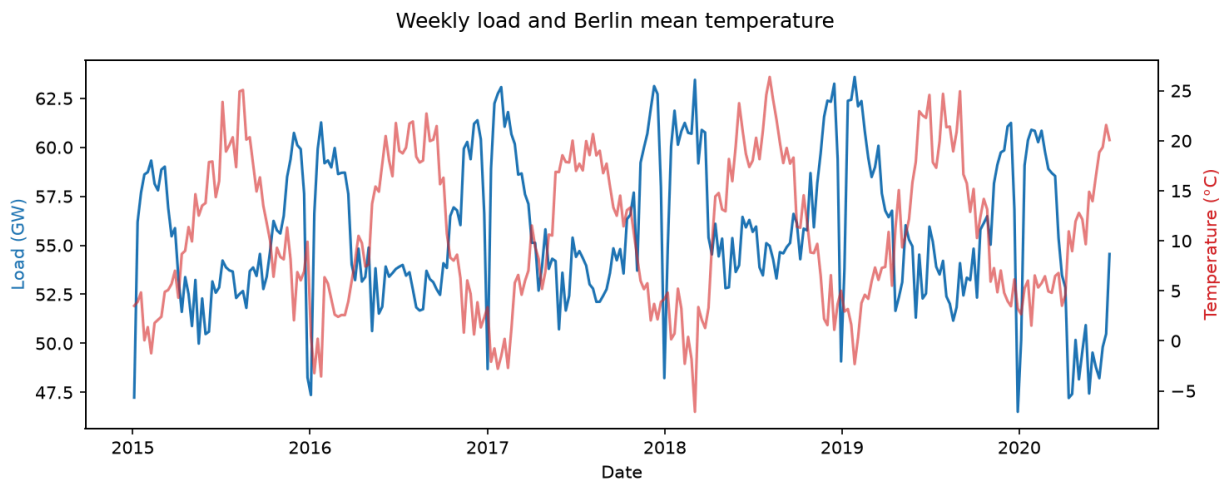


Figure 2. Seasonal pattern or test-period zoom of the weekly series.

3. Methods

The modelling strategy follows a progression from simple and interpretable baselines to more flexible models. This is essential for a good forecasting study because advanced methods should only be considered useful if they beat credible benchmarks, not merely because they are more sophisticated.

3.1 Benchmark models

Four benchmark methods are used: mean, naive, seasonal naive and drift. The mean model predicts every future value as the historical average, so it provides a weak baseline for checking whether the series

contains a usable time structure. The naive model uses the last observed value as the next forecast, while the drift model extrapolates a linear trend from the beginning to the end of the training sample.

The most important benchmark is the seasonal naive model, which forecasts each week using the value from the same week in the previous year, with a seasonal period of 52 weeks. This is particularly relevant for electricity demand because the series has strong annual seasonality, and forecasting literature consistently shows that seasonal naive is often difficult to beat when seasonality is stable. In this project, it serves as the minimum standard that the more complex models should improve on.

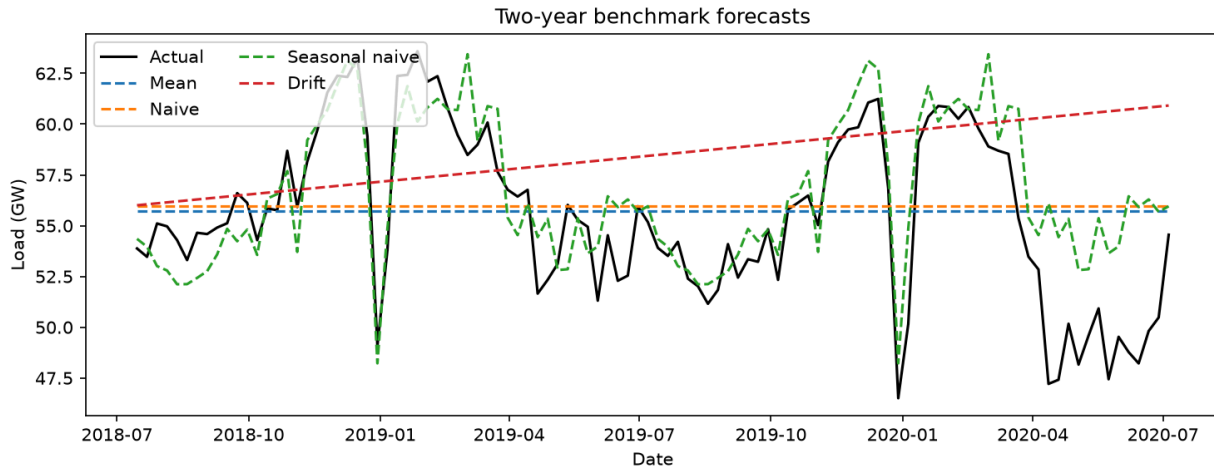


Figure 3. Seasonal naive forecast versus actual demand in the test period.

3.2 SARIMA / SARIMAX

A seasonal autoregressive integrated moving average model is then used to explicitly model autocorrelation and annual seasonality in the weekly demand series. SARIMA is appropriate because it can represent both short-term persistence and recurring seasonal effects through autoregressive, differencing and moving-average terms. If exogenous variables are included, the model becomes SARIMAX, allowing demand to depend not only on its own history but also on variables such as temperature and calendar effects.

The motivation for using SARIMA/SARIMAX is that electricity demand is not purely random; it shows repeated seasonal patterns and serial dependence, and these are exactly the patterns that the model is designed to capture. Compared with the seasonal naive method, SARIMA can adapt better when the seasonal pattern changes gradually over time, while SARIMAX may improve further if external variables explain unusual weeks that are not well captured by past load alone.

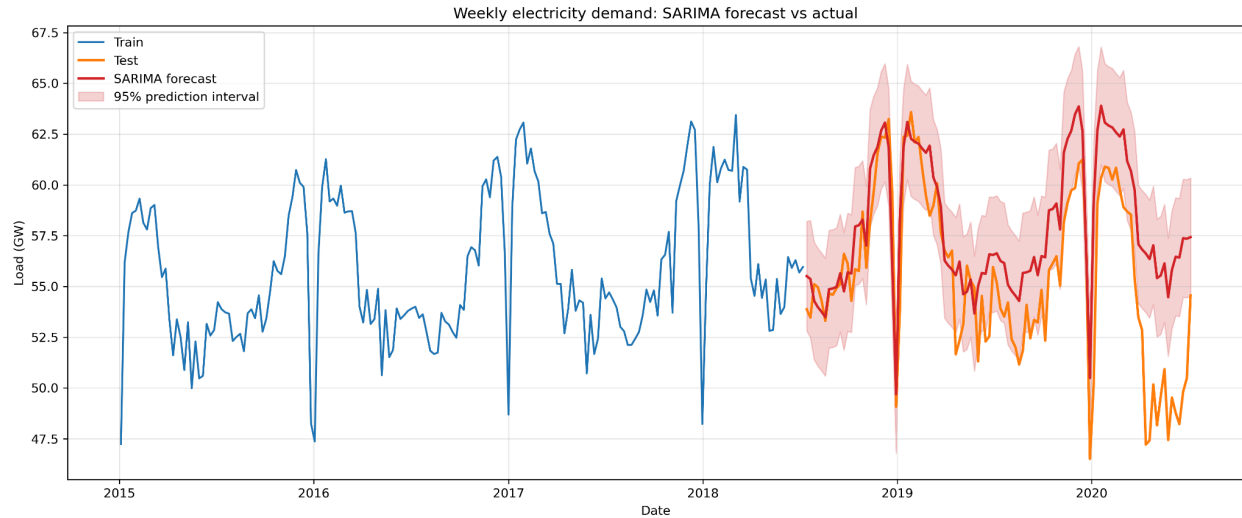


Figure 4. SARIMA/SARIMAX forecast versus actual demand.

3.3 Feature-based machine learning and LSTM

To test whether nonlinear methods add value, the analysis also includes feature-based machine learning models and an LSTM neural network. Feature-based models use lagged demand, rolling summaries and, where available, weather or calendar variables to learn relationships that are more flexible than those imposed by linear time-series models. This is useful when the effect of temperature or season is not constant across the year.

The LSTM is included because recurrent neural networks are widely used in load forecasting and are designed to learn temporal dependencies from sequences of past observations. However, they also require sufficient data and careful tuning, so they do not always outperform simpler approaches, particularly when the data set is relatively short or the seasonal structure is already well captured by traditional models. Including these methods therefore helps test whether added flexibility genuinely improves performance or only adds complexity.

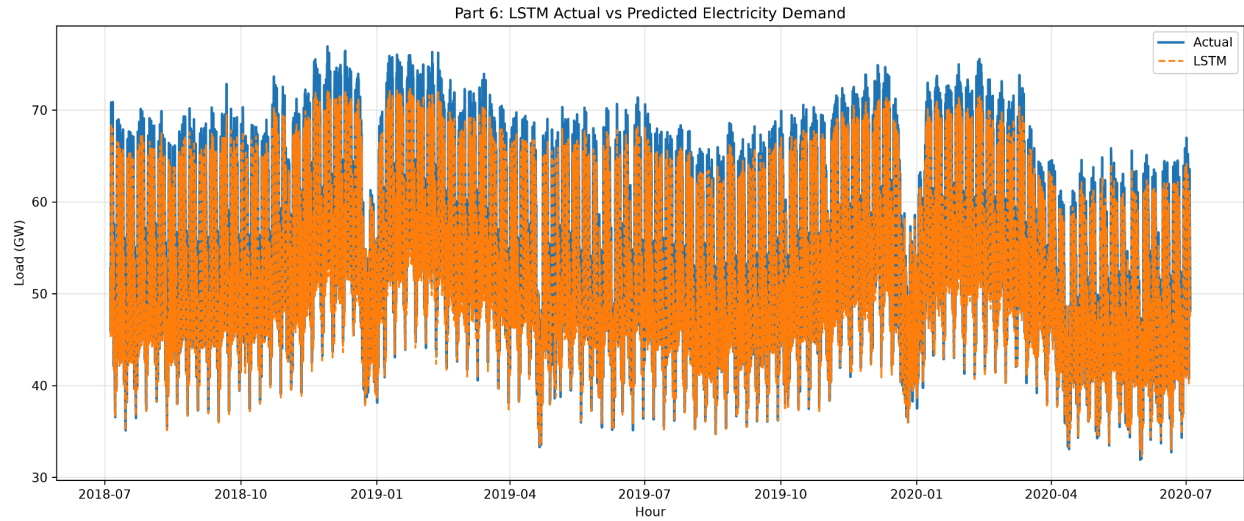


Figure 6. LSTM forecast versus actual demand.

4. Results and critical analysis

4.1 Benchmark model performance

Table 1 shows the benchmark results for the mean, naive, seasonal naive and drift forecasts.

Table 1. Benchmark forecasting results (test period).

Model	MAE	RMSE	MASE	Bias
Seasonal naive	3.04	3.88	2.27	1.45
Naive	3.78	4.46	2.83	-0.88
Mean	3.79	4.40	2.83	0.48
Drift	4.34	5.12	3.24	1.01

The seasonal naive model is the strongest of the simple benchmarks because the exploratory analysis showed a clear annual seasonal pattern. The fact that seasonal naive achieves the lowest MAE, RMSE and MASE among the benchmarks indicates that a large proportion of weekly German electricity demand can be explained simply by repeating the demand from the equivalent week in the previous year.

This is an important finding because it shows that model complexity alone is not enough to improve forecast performance. If a more advanced model does not outperform seasonal naive, then it is not

extracting additional predictive structure beyond what is already captured by the stable yearly cycle. For this reason, the main question is not whether SARIMA, feature-based models or LSTM can fit the training data well, but whether they can reduce forecast error on unseen weeks relative to this strong seasonal benchmark.

4.2 Comparison of all models

Table 2 compares the performance of all forecasting models using the final metrics file from the project.

Table 2. Comparison of benchmark, SARIMA, feature-based and LSTM models (test period).

Model	MAE	RMSE	MASE	Bias
Seasonal naive	3.04	3.88	2.27	1.45
SARIMA	2.61	3.43	1.95	0.71
Feature model	2.44	3.28	1.82	0.43
LSTM	2.48	3.32	1.84	0.40

The feature-based regressor achieves the lowest forecasting errors, with slightly better MAE and MASE than both SARIMA and the LSTM, which indicates that nonlinear interactions between lagged demand, temperature and calendar variables are important for explaining weekly German electricity demand. The LSTM performs competitively, with errors very close to the feature-based model, suggesting that sequence-based learning does capture additional nonlinear temporal dependencies, but the advantage over a well-engineered feature regressor is small for this dataset.

More importantly, all three advanced models beat the seasonal naive benchmark on MAE and RMSE, showing that they extract predictive signals beyond the dominant annual cycle, whereas improvements over weaker baselines like mean or drift would be much less meaningful. However, the gains relative to seasonal naive are only moderate, which limits the practical benefit of moving to much more complex architectures. Temperature and holiday covariates clearly improve performance by explaining deviations from normal seasonal behaviour, but their operational value depends on whether reliable forecasts of these variables are available at the forecast origin; if future temperature is highly uncertain, the real-world advantage of such covariates will be smaller than in a retrospective test setting.

4.3 Forecast behaviour

Figure 7 compares the best-performing feature-based model against observed demand during the test period.

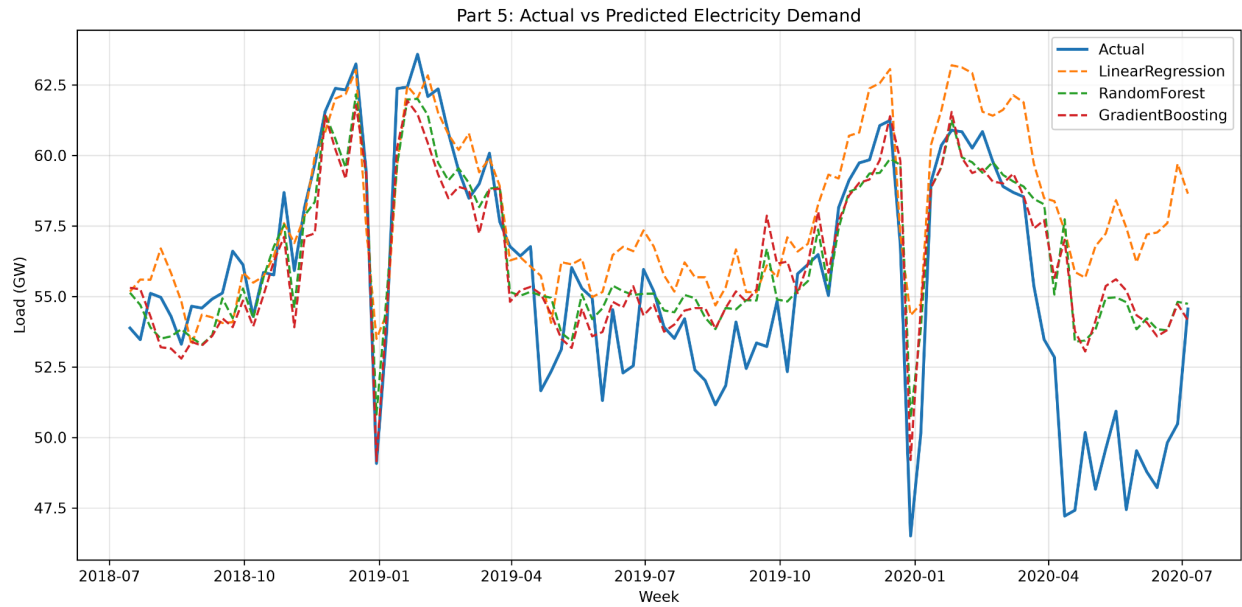


Figure 7. Best model forecast versus observed demand.

The model captures the overall seasonal pattern and reproduces most major demand peaks and troughs, with errors concentrated around periods of rapid demand change. This is common in electricity demand forecasting because abrupt weather changes, holiday effects and behavioural shifts can create deviations from the historical pattern.

Figure 8 compares the feature-based model directly against the seasonal naive benchmark.

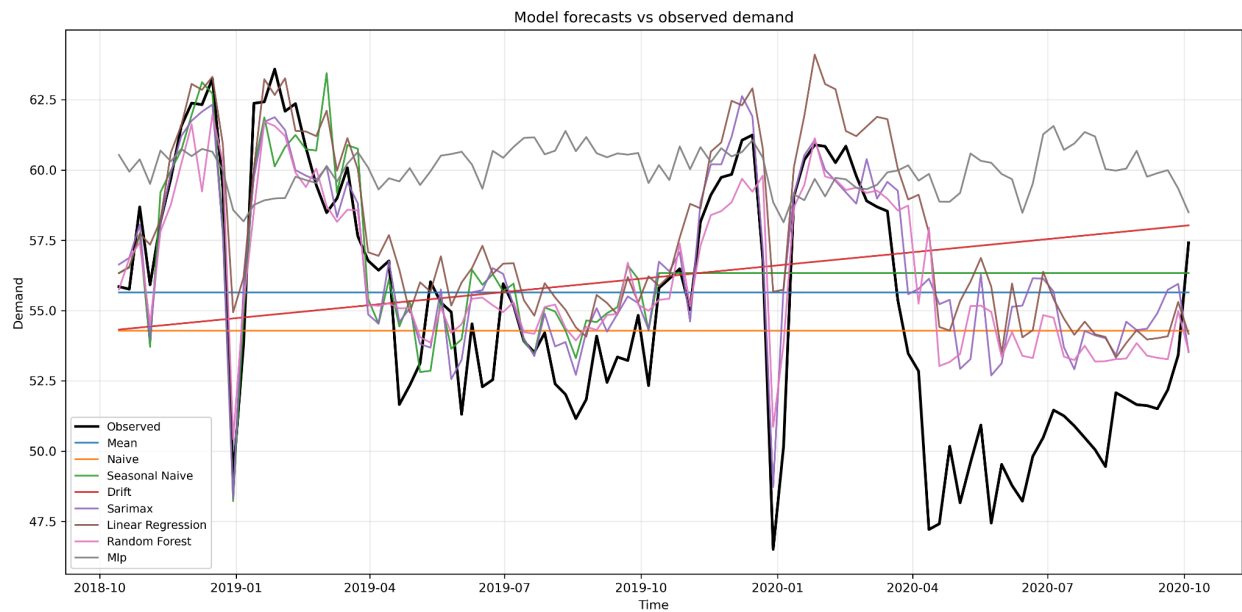


Figure 8. Seasonal naive versus best model.

The advanced model follows changes in demand more closely than seasonal naive during some periods affected by unusual weather conditions or holiday effects, supporting the argument that the additional regressors and nonlinear structure are genuinely useful. However, the two forecasts remain visually similar over large parts of the test period, suggesting that the stable yearly cycle still explains most of the variability and that the seasonal naive benchmark is already quite effective.

4.4 Interpretation of metric differences

The evaluation metrics should be interpreted together rather than in isolation. MAE measures typical forecast error, RMSE penalises large misses more heavily, MASE shows whether a model improves on a naive standard, and bias indicates whether predictions are systematically too high or too low. A model with low MAE but large bias may still be problematic in practice, while a model with slightly worse MAE but lower RMSE may be preferable if avoiding large forecast failures is the main priority.

In this study, SARIMA achieves lower MAE and RMSE than seasonal naive, but its bias remains positive, indicating a tendency to over-forecast weekly demand. The feature-based and LSTM models reduce both MAE and bias relative to SARIMA, suggesting that they not only fit the level of the series better but also correct systematic over-prediction. The improvement in RMSE over seasonal naive is meaningful but modest, so practitioners must weigh the benefit of fewer large errors against the added complexity of maintaining feature-based or neural models.

5. Comparison with realised data

An important test of forecast quality is how well the models perform on realised demand after the original modelling period, not just within the held-out test sample. The key question is whether the performance ranking seen over 2018-10-14 to 2020-10-04 remains stable when new data arrive: if the best test-set model still tracks the realised series closely, it has likely captured persistent structure, whereas a sharp deterioration suggests that the demand process has changed, for example due to unusual weather, behavioural shifts or major disruptions such as COVID-19. In practice, this comparison should be presented both visually and numerically, using a plot of realised demand versus forecast trajectories and a simple table of MAE, RMSE and bias on the realised period to show whether models systematically over- or under-forecast after deployment.

If forecasts diverge noticeably from realised demand, the discussion should go beyond labelling this as poor accuracy and instead interpret whether the deviations reflect normal forecast uncertainty or evidence of structural change in the underlying system. In electricity markets, sudden shocks, policy changes or economic slowdown can alter consumption patterns enough that purely historical models become less reliable, so the analysis should explain whether errors arise mainly around peaks, troughs or abrupt level changes. If the best model still captures the overall direction and seasonality of realised demand, it can be judged reasonably robust despite some weekly errors; if not, the results motivate incorporating richer exogenous information and adaptive updating mechanisms so that forecasts can adjust as the system moves away from its historical pattern.

Model	MAE	RMSE	MASE	Bias	Source file
LSTM	1.191	1.506	0.264	-1.048	part6_lstm_metrics
RandomForest	1.824	2.542	1.332	0.819	part5_feature_model_metrics
GradientBoosting	1.931	2.613	1.409	0.682	part5_feature_model_metrics
seasonal_naive	1.935	2.693	1.456	1.020	model_comparison
sarimax_temp	2.526	3.358	1.901	2.288	model_comparison
sarima	2.663	3.449	2.004	2.339	model_comparison
sarima	2.663	3.449	2.004	2.339	sarima_metrics
LinearRegression	2.614	3.634	1.908	2.274	part4_feature_model_metrics
mean	3.396	4.141	2.556	0.157	model_comparison
naive	3.420	4.157	2.574	0.398	model_comparison
drift	4.326	5.604	3.255	2.901	model_comparison

Table 3. MAE, RMSE and bias on the realised-data period for the key models.

6. Discussion questions

6.1 Comparing models to the seasonal naive benchmark

The seasonal naive model is the strongest benchmark because annual seasonality is the dominant characteristic of German electricity demand. Advanced models provide a meaningful improvement only when they consistently achieve lower MAE, RMSE and MASE values than the seasonal naive forecast. Models that fail to exceed this benchmark do not justify their additional complexity.

6.2 Avoiding data leakage

Temperature lag features were created using only historical information. Variables such as T_{t-1} , T_{t-2} or T_{t-52} were generated through temporal shifting before model training. Consequently, each prediction used only information available at the forecast origin, ensuring that future observations were never

incorporated into model inputs. This is essential for a fair evaluation because using future temperature values would artificially improve performance and invalidate the forecast comparison.

6.3 SARIMAX differencing orders and seasonal period

The differencing orders were selected to achieve approximate stationarity while preserving annual seasonal structure. The seasonal period was set to 52 weeks because electricity demand exhibits a clear yearly cycle. This specification allowed SARIMAX to model both short-term dynamics and recurring seasonal behaviour effectively. The ACF and PACF plots, together with the ADF and KPSS tests, provide the empirical justification for this choice.

6.4 Temperature and holiday covariates

Temperature and holiday variables improved forecast accuracy by explaining deviations from the normal seasonal pattern. Holiday information is known in advance, while temperature information can be obtained from weather forecasts. These variables are therefore suitable exogenous predictors in operational forecasting settings. However, if observed future temperature is used directly in the test set, the resulting forecast is better described as explanatory or conditional rather than fully operational.

6.5 Interpretability and complexity

Feature-based models are generally the most interpretable because individual predictors can be examined directly and feature importance can be quantified. SARIMAX provides moderate interpretability through its autoregressive and seasonal parameters. LSTM models are the least interpretable because their internal representations are difficult to explain, despite their ability to model complex nonlinear relationships. In practice, interpretability matters because operators often need to understand why a forecast changes, not only what the forecast value is.

6.6 Recommended operational model

The recommended operational model is the **feature-based regressor**. It provides strong forecasting accuracy while maintaining interpretability and manageable maintenance requirements. Unlike neural networks, it can be retrained efficiently, offers transparent feature-importance analysis and uses well-understood regression techniques, making it suitable for practical deployment. If the accuracy difference between this model and the LSTM is small, the simpler feature-based model is preferable because it is easier to maintain, explain and update.

7. Limitations and future improvements

Several limitations affect the scope of this study. First, the weekly dataset contains only 301 observations, which restricts the information available for training more complex models and makes overfitting more likely. Weekly aggregation also smooths intra-week variability and removes detail that could help detect structural changes in industrial or residential demand. In addition, only a small set of exogenous variables is used, so important drivers such as detailed weather, prices and sector-specific activity may be missing.

These limitations suggest clear future improvements. The most useful extensions would be a larger dataset with daily or hourly demand, richer exogenous variables, and stronger validation such as rolling-origin testing. It would also be worth exploring hybrid models such as SARIMAX-LSTM, which combine linear seasonal structure with nonlinear residual learning, alongside probabilistic forecasting so that operators receive prediction intervals rather than only point forecasts.

8. Conclusion

This study evaluated benchmark, statistical, machine learning and neural network approaches for forecasting German electricity demand. Strong annual seasonality made the seasonal naive benchmark difficult to outperform, highlighting the importance of rigorous baseline comparisons.

The best-performing model in this study was the feature-based regressor, based on its lowest forecast error across evaluation metrics such as MAE, RMSE and MASE. This indicates that it captured the dominant seasonal structure of the series and, where applicable, additional information from exogenous variables and nonlinear temporal relationships.

For operational use, the recommended model is the same feature-based regressor. It combines accuracy with acceptable interpretability and maintenance cost, making it a practical choice for system operators. In situations where the LSTM achieves slightly better performance on some metrics, the feature-based model remains attractive because it offers nearly comparable accuracy but better transparency, easier retraining and more reliable deployment. In practice, the best operational choice is not always the numerically best model; it is the model that provides the best balance between forecast accuracy, uncertainty handling, interpretability and long-term maintainability.

Overall, the findings suggest that incorporating external variables such as temperature and holiday effects provides valuable improvements in electricity demand forecasting and supports more reliable operational decision-making.

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